Representational Dynamics and Infrared Rigidity: A Constructive Research Programme for the Yang–Mills Mass Gap

Research dossier

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Abstract

We formulate a representation-aware programme for the four-dimensional Yang–Mills existence and mass-gap problem. The physical theory is treated as coherent information transported among regulated measures, observable algebras, renormalization-group descriptions, Euclidean correlation functionals, and reconstructed spectral data. We define uniform physical relational rigidity, prove a positive spectral transfer theorem, give an abstract noncircular influence-basin theorem, type the required representation morphisms, and exhibit counterexamples that invalidate several common shortcuts. We then isolate the first genuinely open four-dimensional step as a cutoff- and volume-uniform full-measure basin-entry inequality. The result is a rigorous conditional architecture and a falsifiable research programme, not a solution of the Clay problem. Its strongest claim level is C3: a conditional continuum/transfer theorem.

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1 Epistemic statement and exact target

The Clay problem asks for a nontrivial quantum Yang–Mills theory on $\mathbb{R}^4$ for every compact simple gauge group G, satisfying axioms at least as strong as those conventionally required for a relativistic quantum field theory, and possessing a mass gap $\Delta > 0$ above the vacuum [1]. This dossier does not prove that theorem. It separates the theorem into interfaces so that a proof cannot move spectral information through a representation change without declaring what is preserved.

Definition 1.1 (Clay target)**.** *For a compact simple Lie group G, a* Clay-level solution *supplies a physical Hilbert space $\mathcal{H}_G$, a unit vacuum Ω_G, a positive self-adjoint Hamiltonian H_G, a nontrivial gauge-invariant local quantum field theory satisfying the required relativistic axioms, and a number $\Delta_G > 0$ such that*

$$\operatorname{spec}(H_G) \cap (0, \Delta_G) = \varnothing,$$

with nonvacuum finite-energy content. The construction must be obtained from mathematically controlled data, not by postulating the Hamiltonian or gap.

The mass gap is distinct from confinement, an area law, a one-particle pole, or a gap for an auxiliary stochastic generator. Any one of these may be related to a physical gap under additional hypotheses; none is a synonym.

1.1 Claim hierarchy

Level	Meaning
C0	concepts and definitions
C1	finite or executable calibration
C2	rigorous restricted-model theorem
C3	conditional continuum or representation-transfer theorem
C4	regulator-specific nonperturbative construction, uniform limits incomplete
C5	continuum construction and gap for a restricted target
C6	the full Clay target

Nothing proved below exceeds C3. This ceiling is a mathematical conclusion of the dependency audit, not a rhetorical disclaimer.

2 Ontology: a universe as coherent representational dynamics

The proposed ontology does not make representation more real than physics. It says instead that physically meaningful identity is the invariant content of a coherently related family of partial descriptions. A universe is represented by

$$\mathfrak{U} = (\mathbf{Rep}, \mathbf{Mor}, \mathcal{I}, \mathcal{P}, \mathcal{L}),$$

where **Rep** is a class of descriptions, **Mor** contains admissible transformations among descriptions, $\mathcal{I}$ assigns invariant physical content, $\mathcal{P}$ records preservation certificates, and $\mathcal{L}$ specifies limiting coherence. No individual regulator is the universe. The universe is the compatible structure that remains identifiable through admissible changes of resolution and language.

In this interpretation:

- gauge potentials are coordinates on relational physical information;
- regulators are temporary realizations that expose different proof structures;
- renormalization is a controlled transformation of descriptive scale;
- particles are spectral interpretations in a reconstructed representation;
- the continuum theory is a coherence condition across limiting representations; and
- a mass gap is invariant infrared rigidity: sufficiently low positive energy cannot be created by changing a faithful physical description.

The ontology becomes mathematical only after the words "controlled", "faithful", and "invariant" are replaced by typed maps, topologies, error bounds, and density theorems.

3 Physical observables and completeness

Let $\Lambda_{a,L}$ be a regulated spacetime with cutoff a and physical linear size L. Its kinematical fields form $\mathcal{K}_{a,L}$ and its gauge group is $\mathcal{G}_{a,L}$. The primary physical algebra is

$$\mathcal{A}_{\mathrm{phys},a,L} = \{F : \mathcal{K}_{a,L} \to \mathbb{C} : F(g \cdot A) = F(A) \text{ for all } g \in \mathcal{G}_{a,L}\},$$

completed only after a state and topology have been specified.

Candidate cores include local curvature composites after renormalization, smeared gauge-invariant operators, Wilson loops, and extended operators needed to distinguish global or topological sectors. Wilson loops alone are not assumed complete.

Definition 3.1 (State-separating core)**.** *A $*$-subalgebra $\mathcal{D} \subset \mathcal{A}_{\mathrm{phys}}$ is state separating if equality of all expectation values on $\mathcal{D}$ implies equality of physical states. After OS reconstruction it is* spectrally complete *if*

$$\overline{\{A\Omega : A \in \mathcal{D}\}} = \mathcal{H}.$$

For a mass-gap argument it suffices to require density of centered vectors in $\Omega^{\perp}$.

Frontier obligation 3.2 (Observable completeness). Prove that the regulated observable cores converge to a continuum core whose centered vectors are dense in the reconstructed vacuum complement. Until then, any correlation result is sectorial.

4 Representation packages and typed morphisms

Definition 4.1 (Representation package)**.** *A representation is a tuple*

$$\mathsf{R} = (\mathcal{K}, \mathcal{A}_{\mathrm{phys}}, \mathcal{S}, \mathcal{D}, \mathcal{G}, \mathcal{L}, \mathcal{Q}),$$

consisting of kinematics, physical observables, states or measures, dynamics, gauge structure, locality/scale data, and quantitative control data.

A morphism $F : \mathsf{R} \to \mathsf{R}'$ contains maps on states and observables with the dual compatibility relation

$$\langle F_* \omega, A' \rangle \simeq \langle \omega, F^* A' \rangle,$$

where $\simeq$ is equality for exact maps and an explicitly bounded error for approximate maps. Every morphism carries a certificate

$$\mathrm{cert}(F) = (g, p, \ell, o, u),$$

recording gauge compatibility, reflection positivity, locality, observable completeness, and uniformity in volume.

Proposition 4.2 (Composition ledger PROVED)**.** *Let $F : \mathsf{R}_0 \to \mathsf{R}_1$ and $G : \mathsf{R}_1 \to \mathsf{R}_2$ have additive expectation errors ε_F and ε_G. Then $G \circ F$ has error at most $\varepsilon_F + \varepsilon_G$, and*

$$\mathrm{cert}(G \circ F) = \mathrm{cert}(F) \wedge \mathrm{cert}(G)$$

componentwise. Thus a lost certificate cannot reappear without a separate recovery theorem.

Proof. Insert and subtract the intermediate expectation and use the triangle inequality. A property claimed for the composition must hold at both stages; otherwise the corresponding Boolean certificate is false. The executable artifact checks this rule with exact rational errors. □

The important morphism classes are exact equivalence, regulator embedding, coarse graining, renormalized observable transport, gauge fixing with global residual data, reconstruction, and continuum limiting. Calling every one of them an "equivalence" would erase precisely the information the programme is designed to audit.

5 Regulator atlas

The programme is not founded on earlier Wilson-action work. The Wilson action appears as one regulator among several because any final continuum conclusion needs a regulator choice or a regulator-comparison theorem.

Regulator	Gauge invariance	Positivity/locality	Principal proof obligations
Wilson plaquette	exact lattice gauge invariance	local; standard reflection-positive constructions exist in suitable settings	weak-coupling continuum trajectory, full-measure RG, physical scaling, universality
Heat-kernel plaquette	exact class-function gauge invariance	local; positivity must be checked for the chosen time slicing	coefficient control, comparison with continuum renormalization, universality
Villain-type or auxiliary-field formulations	gauge invariance depends on nonabelian implementation	may expose convexity but positivity is formulation dependent	auxiliary-field erasure, topology, equivalence to target theory

Regulator	Gauge invariance	Positivity/locality	Principal proof obligations
Hamiltonian lattice gauge theory	Gauss law and physical Hilbert space explicit	spectral interpretation native; Euclidean covariance indirect	continuum limit, locality, Lorentz covariance, comparison with Euclidean measures
Gradient-flowed observables over another regulator	gauge-covariant smoothing of observables	improves ultraviolet observables but is not a stand-alone theory	invertibility/sufficiency of flowed data, zero-flow-time limit

Balaban's work establishes deep multiscale control for lattice gauge theories, including a four-dimensional small-field coupling renormalization analysis [4]. It is indispensable evidence for the feasibility of exact RG methods, but it does not supply the complete full-measure, continuum, OS-reconstruction, and spectral chain required here.

6 Uniform physical relational rigidity

Let $a_n \downarrow 0$, $L_n \uparrow \infty$, and let ω_n be regulated states. Let $\tau_t^{(n)}$ denote Euclidean time translation in physical units, with interpolation when t is not a lattice multiple. For a centered observable A_n, write

$$C_n(A_n; t) = \omega_n(A_n^* \tau_t^{(n)} A_n) - |\omega_n(A_n)|^2.$$

Definition 6.1 (Uniform physical relational rigidity)**.** *The family has UPRR with rate $m > 0$ on a continuum core $\mathcal{D}$ if every $A \in \mathcal{D}$ has regulated representatives A_n, a constant $K_A < \infty$ independent of n and L_n, and errors $r_n(A, t)$ such that*

$$\begin{aligned} 0 \le C_n(A_n; t) &\le K_A \mathrm{e}^{-mt} + r_n(A, t), \\ \lim_{n\to\infty} r_n(A, t) &= 0 \quad \textit{for every fixed } t \ge 0, \end{aligned} \tag{1}$$

and the quadratic forms converge to continuum correlations. The parameter m has units of inverse physical length. Polarization supplies mixed correlators when required.

The definition is noncircular only if (1) is obtained from an independent local or multiscale criterion. It must not be used as the definition of the RG basin whose entry one then claims to prove.

For a one-step transfer eigenvalue λ_a, the dimensionless gap is $\delta_a = -\log \lambda_a$ while the physical energy is δ_a/a. A lower bound on δ_a that shrinks faster than a is irrelevant to the continuum mass gap.

7 Positive spectral transfer

Lemma 7.1 (Positive Laplace support lemma PROVED)**.** *Let μ be a finite positive measure on $[0, \infty)$ and*

$$C(t) = \int_{[0,\infty)} \mathrm{e}^{-t\lambda}\, d\mu(\lambda).$$

If $C(t) \le K\mathrm{e}^{-mt}$ for all $t \ge 0$, then $\mu([0, m)) = 0$.

Proof. For $0 < \varepsilon < m$,

$$\mu([0, m-\varepsilon])\mathrm{e}^{-t(m-\varepsilon)} \le C(t) \le K\mathrm{e}^{-mt}.$$

Thus $\mu([0, m-\varepsilon]) \le K\mathrm{e}^{-\varepsilon t}$ for every t, and the left side is zero. Let $\varepsilon \downarrow 0$ through a countable sequence. □

Theorem 7.2 (Dense-core rigidity transfer PROVED)**.** *Let $H \ge 0$ be self-adjoint on $\mathcal{H}$, with unique vacuum Ω, and let $\mathcal{D}_0 \subset \Omega^\perp$ be dense. Suppose for every $\psi \in \mathcal{D}_0$ there is $K_\psi < \infty$ such that*

$$\langle \psi, \mathrm{e}^{-tH}\psi\rangle \le K_\psi \mathrm{e}^{-mt} \qquad (t \ge 0).$$

Then

$$\operatorname{spec}(H|_{\Omega^\perp}) \subset [m, \infty).$$

Proof. For $\psi \in \mathcal{D}_0$, the spectral theorem gives a positive measure μ_ψ and Lemma 7.1 shows $\|E_H([0,m))\psi\|^2 = 0$. The spectral projection is bounded, so density gives $E_H([0,m)) = 0$ on $\Omega^\perp$. □

Theorem 7.3 (Regulator-to-continuum transfer CONDITIONAL)**.** *Assume regulated representatives of a continuum observable core satisfy UPRR, their centered quadratic forms converge for every fixed t, reflection positivity reconstructs e^{-tH}, and their continuum vectors are dense in $\Omega^\perp$. Then H has no spectrum in $(0, m)$.*

Proof. Take $n \to \infty$ in (1) at fixed t to obtain the hypothesis of Theorem 7.2 on the dense continuum core. Apply that theorem. □

Fixed-time error convergence is enough for this conditional theorem because the continuum estimate is established separately for every t. It is not enough to infer a uniform regulated spectral gap if the large-time and cutoff limits are interchanged before a continuum semigroup exists.

8 A noncircular infrared basin

We now give an abstract basin whose membership is local in a block specification rather than defined by the desired global correlation decay. Let I index blocks with metric d. For a local observable F, let $\delta_i(F)$ be its oscillation under a change of block i. Let $C = (c_{ij})$ be a nonnegative influence matrix derived from conditional kernels.

Definition 8.1 (Weighted influence basin)**.** *For $\kappa > 0$ and $q < 1$, an effective theory lies in $\mathfrak{B}_{\mathrm{rig}}(\kappa, q)$ if its conditional specification obeys the comparison estimate*

$$|\operatorname{Cov}(F,G)| \le K_0 \sum_{i,j} \delta_i(F) D_{ij} \delta_j(G), \qquad D = (I-C)^{-1}, \tag{2}$$

and

$$\|C\|_\kappa := \sup_i \sum_j \mathrm{e}^{\kappa d(i,j)} c_{ij} \le q < 1. \tag{3}$$

All constants must be uniform in the total volume. In a concrete Gibbs specification, (2) is to be proved by a Dobrushin comparison argument; it is displayed rather than hidden inside the word "mixing".

Theorem 8.2 (Basin-to-clustering CONDITIONAL)**.** *If an effective theory lies in $\mathfrak{B}_{\mathrm{rig}}(\kappa, q)$ and F, G have finite block supports S_F, S_G, then*

$$|\operatorname{Cov}(F,G)| \le \frac{K_0}{1-q}\, \mathrm{e}^{-\kappa d(S_F, S_G)} \Big(\sum_i \delta_i(F)\Big)\Big(\sum_j \delta_j(G)\Big).$$

Proof. The triangle inequality implies that the weighted row norm is submultiplicative, so $D = \sum_{n\geq 0} C^n$ and $\|D\|_\kappa \leq (1-q)^{-1}$. Consequently $D_{ij} \leq (1-q)^{-1} \mathrm{e}^{-\kappa d(i,j)}$. Insert this in (2) and use $d(i,j) \geq d(S_F, S_G)$ for nonzero oscillations. □

Corollary 8.3 (RG scale to physical rigidity CONDITIONAL)**.** *Suppose a gauge-compatible RG flow reaches $\mathfrak{B}_{\mathrm{rig}}(\kappa, q)$ at block length ℓ_*; observable pullback has summable errors and uniformly controlled support growth. Then the original correlations decay with physical rate at least*

$$m_{\mathrm{corr}} = \frac{\kappa}{\ell_*},$$

up to the declared pullback errors. If the controlled direction is Euclidean time and OS hypotheses hold, Theorem 7.3 yields the same lower bound for the Hamiltonian gap.

This result separates two questions: the implication from basin membership is proved, while membership of the four-dimensional Yang–Mills trajectory is not.

9 Gauge-covariant RG and the basin-entry frontier

A complete RG step must transform the measure, action, observables, scale, and reconstruction data together. Schematically,

$$(\mu_k, S_k, \mathcal{A}_{\mathrm{phys},k}, Z_k, \ell_k) \longmapsto (\mu_{k+1}, S_{k+1}, \mathcal{A}_{\mathrm{phys},k+1}, Z_{k+1}, b\ell_k).$$

It must report gauge covariance, locality or quasilocality, reflection positivity, volume uniformity, running couplings, irrelevant remainders, and errors. A perturbative beta function alone is not this map.

The large-field decomposition must be gauge invariant, for example through plaquette-curvature or orbit-distance events. Exceptional blocks require activities summable over positions, polymers, and RG scales. Global holonomy, center flux, nontrivial bundles, and instanton-like sectors must either be included or bounded. A coordinate chart near the trivial connection does not cover these obligations.

Definition 9.1 (Basin-entry statement BE)**.** $\mathrm{BE}(k_*, \kappa, q)$ *is the assertion that, along the cutoff trajectory defining four-dimensional pure Yang–Mills, there is a finite physical entry scale $\ell_* = b^{k_*} a$ such that the complete effective conditional kernels $\gamma_B^{(k_*)}$ have influence coefficients satisfying*

$$\sup_B \sum_C \mathrm{e}^{\kappa d(B,C)} c_{BC}^{(k_*)} \leq q < 1, \tag{4}$$

and satisfy the comparison estimate (2)*, with all constants uniform in volume and cutoff and all reconstruction errors summable.*

Frontier obligation 9.2 (Central frontier OPEN)*.* Prove $\mathrm{BE}(k_*, \kappa, q)$ for the full four-dimensional pure Yang–Mills measure, or prove an equally explicit noncircular polymer, coercivity, or transfer-contraction criterion. No such proof is supplied here.

Asymptotic freedom controls the ultraviolet running of the coupling but does not imply (4). Strong-coupling lattice theorems, such as exponential correlation decay established by functional-inequality methods in explicit parameter regimes [5, 6], are rigorous restricted calibrations. They do not by themselves connect the weak-coupling continuum trajectory to a positive physical gap.

9.1 Concrete research sub-obligations

BE1. Derive a gauge-orbit block specification with quantitatively local conditional kernels.

BE2. Establish weak-coupling small-field bounds with constants stable for the required number of RG steps.

BE3. Prove a summable exceptional-polymer estimate uniform in volume.

BE4. Cover global orbit and topological sectors and control atlas overlaps.

BE5. Bound generated nonlocal interactions in a weighted norm strong enough for (4).

BE6. Show that the flow enters $q < 1$ at finite physical ℓ_* rather than merely at a cutoff-dependent scale.

BE7. Preserve or recover reflection positivity and retain a spectrally complete observable transport.

These are detailed todos in the literal mathematical sense: each names a finite theorem or inequality whose proof can be accepted or rejected.

10 Continuum coherence

Let $S_n^{(a,L)}$ denote renormalized n-point Schwinger functionals on a gauge-invariant test algebra. A continuum theory requires more than a formal limit of actions.

Definition 10.1 (Continuum coherence package)**.** *A package consists of:*

(i) a topology in which all $S_n^{(a,L)}$ converge;

(ii) Euclidean covariance, symmetry, locality, reflection positivity, and the regularity conditions required by the chosen OS theorem;

(iii) nontriviality and a certificate identifying the limit as pure Yang–Mills under stated renormalization conditions;

(iv) uniqueness or a regulator-comparison theorem in a declared class; and

(v) justified interchange of cutoff, volume, RG-depth, observable, and large-separation limits.

The original OS programme reconstructs relativistic quantum theory from suitable Euclidean Green functions [2, 3]. The precise regularity hypotheses matter: the corrected second paper is not interchangeable with an informal list of covariance and positivity properties.

Theorem 10.2 (Conditional continuum gap theorem CONDITIONAL)**.** *Assume a continuum coherence package for nontrivial pure Yang–Mills, UPRR with rate $m > 0$ on an OS-dense gauge-invariant core, a unique vacuum, and nonempty finite-energy nonvacuum content. Then the reconstructed Hamiltonian obeys*

$$\operatorname{spec}(H) = \{0\} \cup \Sigma, \qquad \Sigma \subset [m, \infty), \quad \Sigma \neq \varnothing.$$

Proof. The coherence package and OS theorem construct $\mathcal{H}, \Omega, H$. UPRR passes to the continuum by Theorem 7.3. Density upgrades the sectorial conclusion to all of $\Omega^{\perp}$, and the final assumption gives $\Sigma \neq \varnothing$. □

Frontier obligation 10.3 (Continuum construction OPEN). Construct the above package in four dimensions with uniform estimates strong enough to pass UPRR, and prove regulator independence. This dossier does not solve that constructive field-theory problem.

11 Counterexample bank

Counterexample 11.1 (Finite-volume fake gap)**.** *Let $H_L = \mathrm{diag}(0, L^{-1})$. Every finite L has a positive gap, but the gaps converge to zero. Positivity without volume uniformity proves no infinite-volume gap.*

Counterexample 11.2 (Lattice-unit collapse)**.** *Let $a_n = n^{-1}$ and let a transfer operator have dimensionless gap $\delta_n = a_n^2$. Then $\delta_n > 0$ for every regulator, while the physical energy is $\delta_n/a_n = a_n \to 0$.*

Counterexample 11.3 (Hidden gapless sector)**.** *Take*

$$\mathcal{H} = \mathbb{C}\Omega \oplus \mathbb{C}g \oplus L^2([0,1]), \qquad H = 0 \oplus 1 \oplus M_\lambda.$$

An observable class whose centered vectors span only $\mathbb{C}g$ has exact decay e^{-t}, yet H has spectrum arbitrarily close to zero in the hidden L^2 sector. Observable density is indispensable.

Counterexample 11.4 (Why positivity matters)**.** *If signed spectral weights are permitted, low-energy terms can cancel. For example $\delta_\lambda - \delta_\lambda$ has zero Laplace transform while being formally supported at any chosen low energy. Reflection positivity is what turns Euclidean quadratic correlations into positive spectral measures.*

Counterexample 11.5 (Stochastic gap is not physical energy)**.** *For a Markov generator L with invariant measure μ, replacing L by cL leaves μ unchanged and multiplies its Poincaré spectral gap by c. The Euclidean quantum theory determined by μ has not thereby acquired an arbitrarily large physical mass. A comparison theorem must link the sampling dynamics to physical time before its mixing rate has spectral meaning.*

Counterexample 11.6 (A vacuous local sector)**.** *A topological theory or an observable subalgebra with only multiples of the identity can satisfy any centered local correlation estimate vacuously. The Clay target also requires a nontrivial theory and physical nonvacuum content.*

12 Calibration matrix

Model	UPRR	Complete core	Physical gap	Verdict
finite diagonal $H = \mathrm{diag}(0,1,2)$	yes	yes	yes	accepts
massive Gaussian field	yes	standard field core	yes	accepts
massless Gaussian field	no uniform $m > 0$	yes	no	rejects
$H_L = \mathrm{diag}(0, L^{-1})$	nonuniform	yes	no limit gap	rejects
hidden gapless direct sum	sectorial yes	no	no	rejects
topological/trivial local sector	vacuous	generally no	not sufficient	rejects
strong-coupling lattice YM	restricted decay	regulator core	lattice result	calibration only
four-dimensional continuum YM	desired	open	open	frontier

The finite diagonal, nonuniform, hidden-sector, positivity, and morphism-error cases are implemented in `finite_spectral_core.py` and its unit tests. Because no proof assistant is installed in the execution environment, the artifact uses exact rational arithmetic and fail-closed runtime validation. It is an executable C1 certificate, not formal verification of the analytic frontier.

13 Limit-order ledger

The following implications require named interchange theorems:

$L \to \infty$	thermodynamic limit,
$a \to 0$	continuum limit,
$t \to \infty$	spectral extraction,
$k \to k_*(a)$	RG iteration to a physical scale,
$A_n \to A$	renormalized observable approximation.

The safe route in Theorem 7.3 takes the regulated limits at each fixed physical t, establishes a continuum positive semigroup estimate for all t, and only then takes $t \to \infty$ inside the spectral lemma. Directly choosing $t = t(a) \to \infty$ requires a stronger uniform error theorem.

14 End-to-end theorem graph

The proposed proof chain is

$$\text{regulated gauge theory} \xrightarrow{\text{typed RG}} \mathrm{BE}(k_*, \kappa, q) \xrightarrow{\text{Thm. 8.2}} \mathrm{UPRR}$$
$$\xrightarrow{\text{continuum coherence}} \text{positive continuum correlations} \xrightarrow{\text{OS+density}} \mathrm{spec}(H|_{\Omega^\perp}) \subset [m, \infty).$$

The middle implications are conditional theorems. The first arrow to basin entry and the continuum-coherence arrow are open in the required four- dimensional strength.

15 Detailed implementation phases and acceptance gates

Phase	Implemented output	Gate	Current status
0	exact target, distinctions, C0–C6 hierarchy	no ambiguity in theorem	PROVED
1	observable taxonomy and density obligation	state separation proved	OPEN
2	representation packages, morphisms, certificates	composition/error theorem	PROVED
3	regulator atlas independent of prior Wilson work	one admissible route selected	CONDITIONAL
4	UPRR definition in physical units	false-positive suite passed	PROVED as definition
5	positive spectral and regulated transfer theorems	positivity+density supplied	CONDITIONAL
6	typed gauge-covariant RG architecture	full step estimates	OPEN
7	gauge-invariant large-field/topology obligations	summable global control	OPEN
8	weighted influence basin	local noncircular criterion	PROVED abstractly
9	basin-to-rigidity theorem	physical rate and pullback	CONDITIONAL
10	four-dimensional basin entry	Eq. (4) uniform	OPEN

Phase	Implemented output	Gate	Current status
11	continuum Schwinger functions and theory identity	OS, nontrivial, universal	OPEN
12	OS reconstruction and spectral extraction	dense core and unique vacuum	CONDITIONAL
13	restricted-model calibration	massive accepted, false positives rejected	RESTRICTED
14	executable finite core	tests pass	RESTRICTED
15	frontier audit and claim downgrade	first open implication named	PROVED

16 Frontier audit

What has been proved here

- typed representation-error composition;
- the positive Laplace-support lemma and dense-core spectral transfer;
- an abstract weighted-influence basin-to-clustering theorem;
- the conditional logical chain from continuum UPRR and OS density to a Hamiltonian gap;
- counterexamples invalidating finite-volume, lattice-unit, incomplete- sector, signed-measure, and stochastic-time shortcuts;
- finite exact-arithmetic calibration of the spectral and morphism core.

What is conditional

The RG-to-rigidity and continuum-gap theorems are valid under their displayed hypotheses. Those hypotheses are not asserted to be known for four-dimensional pure Yang–Mills.

What remains open

1. cutoff- and volume-uniform full-measure entry into a noncircular rigidity basin along the asymptotically free trajectory;
2. summable control of large fields, global gauge orbit geometry, and topological sectors across all required RG steps;
3. construction, nontriviality, uniqueness, OS regularity, and universality of the four-dimensional continuum Schwinger functions;
4. density of the chosen gauge-invariant observable core in the reconstructed physical Hilbert space.

Therefore the strongest justified claim is C3, not C6. The big-picture value of representational dynamics is that it localizes the unresolved physics: the problem is not merely to "find a gap", but to generate an independently verifiable infrared contraction and transport it without losing gauge invariance, positivity, physical scaling, state completeness, or continuum identity.

17 Next theorem-level work

The next productive work is not another rephrasing of the final spectral statement. It is to choose one regulator whose conditional block kernels can be defined globally and attack Eq. (4). Three falsifiable branches are available:

1. *Influence branch:* prove a weighted conditional influence norm below one after a finite physical number of RG steps.

2. *Polymer branch:* derive a convergent full-measure polymer expansion at the entry scale whose connected activities yield temporal clustering.

3. *Transfer branch:* prove a gauge-physical transfer-operator contraction uniform in volume and cutoff, then compare it to UPRR.

Each branch must include a large-field/topology lemma and a positivity/density certificate. Failure of one branch should yield a counterexample or an exact lower-level obstruction, not be hidden by moving to the next representation.

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